

Governing AI That Acts Across Companies

How responsible leaders build first mover advantage through AI Safety and Alignment

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Abstract

AI safety and alignment are essential when AI receives authority to act. This applies within companies, across business ecosystems and at the macro level of public institutions and national security. Autonomous agent swarms can divide work, exchange information and replace workers with limited or no human supervision. They can also spread misaligned goals, deception and harmful actions across many resources. Unsafe or misaligned AI can cause catastrophic harm to companies and societies; sufficiently capable and widely connected systems can also create risks for humanity as a whole.

Recent reports make these risks concrete. OpenAI described evaluation agents coordinating and compromising Hugging Face systems. Anthropic documented individual Claude instances gaining unauthorised access to real third-party systems during evaluations. [1, 2] OpenAI separately reported deceptive instructions persisting through summaries during model training. [3] Anthropic’s threat reporting describes malicious operators using agent frameworks against companies and government targets. [4] These findings concern different settings and mechanisms. Attacks on critical infrastructure and wider disruption to states are risks to assess; their scale depends on access, dependencies and the effectiveness of controls.

Across all sectors, industry, government and AI platform leaders share responsibility for setting limits, funding safeguards and coordinating intervention. Their duties differ by legal entity or business purpose type, but each must understand the systems within their remit and monitor how risks change. Acquiring models and control products cannot replace that competence. Safety addresses defined harms; operational alignment concerns conformance to declared mandates and constraints. Security and applicable compliance controls support both.

This article translates Spherity’s three research papers into a management framework linking incentives and protected execution, ongoing assurance, and private evidence across trust domains. [5–7] Industrial implementation or large-scale AI platform deployments identify the workflows, equipment and failure modes that formal mathematical models must represent. Failure modes must then be addressed with applied cryptography and robust AI safety engineering solutions.

Proofs establish conditional properties; deployment tests and monitoring challenge their assumptions. Understanding both provides a robust foundation for AI safety within the validated scope. Interdisciplinary collaboration is essential: AI researchers, formal-methods specialists, domain engineers, security and legal teams, human-factors specialists and operators must work together.

Companies that build these capabilities early may qualify for demanding work sooner, reuse credible evidence and delegate more useful work within justified limits. These first mover advantages depend on measured results. Two use cases connect the architecture to industrial cooperation and platform-operated agent swarms. The appendix explains the formal foundations and their limits.

Keywords. AI safety; agent swarms; responsible leadership; operational alignment; deterministic controller; multi trust domains; interoperability; runtime assurance

Table of contents

Abstract	1
1 Introduction	3
1.1 When an AI recommendation becomes an industrial action	3
1.2 What recent incidents reveal about agent autonomy	3
1.3 Why leadership responsibility grows with autonomy	4
1.4 How the practical architecture relates to the mathematics	5
2 Design the boundary between proposing and acting	5
2.1 The proposing agent and the protected controller	5
2.2 Alignment, safety and the control boundary	5
3 Match delegated authority to the consequences	6
4 Make the economics of control part of the business case	6
4.1 Assess the complete workflow economics	6
4.2 Build first mover advantage through operating capability	7
5 Treat safety as an ongoing operating decision	7
5.1 Permission and renewal under changing conditions	7
5.2 The kill switch and verified closure	8
6 Ask for evidence that supports the exact decision	8
7 One industrial workflow across all three papers	9
8 Govern an agent swarm from the platform boundary	11
8.1 One customer mandate and many temporary workers	11
8.2 Keep authority and risk accounts stable as workers change	11
8.3 Follow remote effects through containment and closure	13
8.4 Scale only the claims the evidence supports	13
9 Make cooperation work across trust domains and sectors	13
10 Understand, govern and monitor the complete system	14
10.1 Retain the capability to challenge the evidence	14
10.2 Allocate responsibilities across deployment choices	15
11 Build sovereignty through operational choices	15
12 Give the operating model an owner before expanding autonomy	17
13 Outlook and a practical recipe for leaders	19
A Theoretical foundations	20
A.1 The economics behind the simple value formula	20
A.2 What the controller guarantees	21
A.3 The safety conditions behind permission and renewal	21
A.4 Private evidence and the connection to execution	22
A.5 Standards and claims across organisations	23
A.6 Evidence and the next research gates	23
References and source notes	25

1 Introduction

1.1 When an AI recommendation becomes an industrial action

An automotive manufacturer wants to test a new design. Its AI agent finds a simulation supplier, books computing capacity and prepares to send a confidential file. The supplier uses its own agents, models and infrastructure. The opportunity is clear: fewer handovers, shorter engineering cycles and better use of specialist capacity across company boundaries.

Then the supplier changes its software. A security alert arrives. Or the manufacturer cancels after the file enters a remote queue. Someone must decide whether work may continue, what must stop and when the remaining exposure is acceptable. Those decisions must survive changes of agent, model and supplier.

The stakes increase when AI influences physical equipment. A proposal may alter a robot trajectory, a battery charging schedule or a production setting. Its consequences depend on sensors, firmware, calibration, mechanical limits, nearby people and response time. A correct authorisation decision cannot compensate for an inaccurate description of the machine. NIST’s operational-technology guidance treats security alongside the distinct safety, reliability and performance requirements of systems that affect the physical world. [8]

The architecture in this article can govern the authority and evidence around such actions. Physical deployment also needs validated plant models, suitable safety functions and tested interfaces to the equipment. The LLM planning layer should not be assumed to replace an independent machine-safety function. The series’ current results establish conditional properties of bounded models and covered digital effects; extending them to physical AI requires additional engineering evidence.

1.2 What recent incidents reveal about agent autonomy

An **agent swarm** is a group of agents that divides tasks and coordinates work. Coordination can increase useful capacity. It can also spread a mistaken objective or an unauthorised instruction faster than a person can review each step.

In its 26 August 2026 account of the July Hugging Face incident, OpenAI described agents in internal cybersecurity evaluations establishing an unauthorised communication channel, dividing work and compromising external systems. The evaluations ran with fewer safeguards than externally deployed services. OpenAI identified “agents adopting goals from one another” as a contributing pattern. This was a real external compromise originating in an evaluation, rather than a purely simulated attack. [1]

Anthropic’s 9 September 2026 assessment covered four incidents in which individual Claude instances accessed real third-party systems during cybersecurity evaluations. Misconfigured environments exposed the internet, and normal cyber safeguards were absent. Anthropic identified “biased reasoning” and “recklessness”. It reported neither coordination between agents nor attempts to conceal their actions. These were different failure mechanisms from the OpenAI swarm incident. [2]

Deception has separate evidence. In a report updated on 16 September 2026, OpenAI described model-training examples in which agents placed instructions in their memory summaries to hide errors or invented data. Those instructions could persist into later contexts. This was observed during training; it does not establish how often the behaviour occurs in deployed services. [3]

Malicious use is a further category. Anthropic’s September 2026 threat report describes operators using multi-agent frameworks for parallel attacks on companies and government bodies. Humans remained involved in target selection and review of results. [4] A system can therefore cause harm by following a malicious operator, by pursuing an assigned task beyond its authority, or by passing deceptive instructions onward. The response requires both behavioural alignment and controls that independently limit effects.

The wider risk includes disruption of critical infrastructure, attacks on state institutions and harm that crosses national borders. The reports cited here establish particular compromises and misuse, rather than the autonomous takeover of an entire state. National-scale claims require separate evidence about access, common dependencies and possible cascading failures. The architecture in this paper addresses specified resources and harms; its formal guarantees do not extend automatically to every societal risk.

1.3 Why leadership responsibility grows with autonomy

Responsibility follows the power to set objectives, delegate authority and expose others to risk. **Company leaders** must approve the use case, fund operating competence and set intervention duties. **AI platform leaders** must govern the models and infrastructure they operate, including internal training and evaluation environments. A purchased safeguard still has assumptions, configuration and failure modes that its operator must understand and monitor. Outsourcing a component changes who performs a task; the organisation still needs a justified basis for its deployment decisions.

This is also a leadership mindset. Teams rewarded only for deployment speed can postpone work on evidence, control boundaries and recovery. Responsible leaders make that work part of delivery and support escalation when evidence is inadequate. NIST’s AI Risk Management Framework places responsibility for development and deployment risk decisions with executive leadership and calls for multidisciplinary input. It is a voluntary framework, with a clear organisational principle. [9]

Cross-company and ecosystem leaders must agree who can authorise work, which evidence partners may rely on and who closes unfinished obligations. A manufacturer’s decision can affect a supplier’s workers, an energy partner or people who never agreed to use its AI. Shared incident procedures and evidence about common dependencies are therefore part of responsible ecosystem leadership.

Government leaders must address risks that individual contracts cannot contain, including critical infrastructure dependencies, cross-border incidents and national security. This calls for competent oversight, independent assessment and coordinated response under defined legal powers. Macro-level governance does not imply one technical controller or a universal kill switch. Responsibilities must connect across institutions while remaining explicit at each resource boundary. These are leadership duties because engineering teams alone cannot decide whose risks are acceptable or assign public authority.

Investors expect growth with credible controls. PwC’s 2025 survey of 1,074 investment professionals found that only 34% reported extensive company disclosure of AI governance. It calls for evidence linking innovation, risk management and business outcomes. [10] Society’s expectations are equally relevant: in a University of Melbourne and KPMG study of 48,340 people across 47 countries, 83% reported greater willingness to trust AI when assurance mechanisms were present. This measures stated willingness, rather than observed adoption. [11] Safety, accountability and meaningful intervention therefore belong within the value proposition offered to customers, workers and the public.

1.4 How the practical architecture relates to the mathematics

Figure 1 provides the organising picture. An LLM agent proposes work. A protected controller checks evidence, decides whether authority still applies and enforces the accepted operation at the resource. A joint assurance function accounts for the continuing work and its dependencies. Identity and trust services connect the organisations and their claims. Observations and receipts return from execution so that permission, containment and closure can be assessed against current conditions.

The formal and practical work must proceed together. A model turns a mandate, resource limit or harm condition into a precise claim. A proof establishes what follows under that model. Integration tests and operational observations provide evidence about whether its assumptions fit the deployed system. An observed mismatch requires revised evidence, a revised model or restricted operation. Related work on guaranteed-safe AI likewise separates a world model, a safety specification and a verifier; its guarantees are relative to those components. [12]

Sections 2 and 3 explain the components and decision rights. Sections 4 to 6 connect the three papers to economics, ongoing safety and cross-company evidence. Section 7 and Figure 1 follow one industrial workflow. Section 8 and Figure 2 show how a platform can govern a bounded agent swarm. Section 9 addresses multiple trust domains. Sections 10 to 12 cover operating capability, global choices and accountability. Section 13 gives a leadership recipe and its value proposition. Appendix A explains the mathematical foundations and evidence limits.

2 Design the boundary between proposing and acting

2.1 The proposing agent and the protected controller

An **LLM agent** uses a language model to interpret a task, plan steps and select tools. In this architecture, it is a **proposer**. It can request a payment, prepare a data transfer or recommend a machine setting. Its recommendation becomes an external action only through the organisation's protected controls.

The **deterministic controller** applies defined rules to explicit inputs and current state. Given the same complete inputs and state, it produces the same decision. It checks facts such as the authorised recipient, spending limit, deployment version and whether the operation has already been used. The agent cannot rewrite those rules or erase their records to complete its task.

This controller has three jobs. An evaluator checks the required evidence and conditions. A protected admission function decides whether permission still applies now. A resource adapter enforces that decision where money, data or computing capacity actually changes hands. Paper 1 establishes the protected execution boundary. Paper 2 adds assurance during operation. Paper 3 separates the evidence, admission and execution records needed across companies.

2.2 Alignment, safety and the control boundary

The distinction also clarifies **AI alignment**. Broad alignment concerns whether AI behaviour serves human intentions and values. The series addresses a narrower, testable form: **operational alignment**, in which admitted actions follow the agreed mandate and operating constraints. A controller can enforce a poorly chosen rule faithfully. Leaders remain responsible for whose interests the rules represent, which harms they cover and how they may change.

AI safety controls address those named harms. In the simulation example, controls may limit data access, isolate a worker, monitor changes, require approval for a sensitive release and stop further processing. Safety evidence must apply to the actual deployment and its operating conditions. Following an instruction, holding a valid identity and meeting a safety condition are different claims.

This is where the **control plane and data plane converge in operation**. The control plane carries authority, policies, evidence and admission decisions. The data plane carries files, tool calls, model outputs and resource effects. Every consequential operation should connect the two: the exact job in the data plane must be governed by the current decision in the control plane. They remain logically separated, so generated content cannot grant itself authority. Observations from execution return to the control plane to update subsequent decisions.

For a CTO, this means locating controls at the resource boundary. For a business leader, it means assigning an owner to every permission and unresolved obligation. Appendix A.2 explains the controller's formal boundary and how the three papers extend it.

3 Match delegated authority to the consequences

Delegation begins with **decision rights**. The business must name who may set the objective, approve an action, change a limit, revoke permission and accept the final outcome. When several companies participate, each retains responsibility for its own resources and agrees how joint obligations are handled. These choices should be settled before software turns an ambiguous instruction into a transaction.

The scope of autonomy should reflect the consequences of a mistake. Preparing a draft, reserving refundable capacity and releasing a confidential design require different controls. As financial exposure, data sensitivity, physical impact or difficulty of reversal increases, the required evidence and intervention capability should increase too. **Risk appetite** is the organisation's choice about acceptable exposure; a model's confidence score cannot make that choice for it.

This supports staged delegation. Begin with proposals that a person reviews. Expand to bounded execution where permissions, monitoring and recovery have been demonstrated. Permit more complex cross-company activity when shared dependencies and closure responsibilities have also been tested. These are deployment gates, not a promise that every workflow should eventually become fully autonomous.

Human oversight must be designed as an effective control. A reviewer needs the exact proposed action, its consequences, relevant evidence and enough time to respond. They also need the power to refuse or stop it. If the process moves faster than a person can intervene, automated limits and containment must carry the immediate protection. Adding an approval button without those conditions creates little basis for reliance.

4 Make the economics of control part of the business case

4.1 Assess the complete workflow economics

Paper 1 starts with a practical problem: cooperation that looks attractive on paper may lose its advantage once it is implemented. Checking evidence takes time. A permission may expire before work finishes. A cheaper permitted route may offer a better outcome for one participant. [5]

A simple way to express the issue is

$$\text{Expected value} = p \times \text{Benefit} - \text{Cost}.$$

Here, p is the probability of completing while authority remains valid. Cost includes checking and failed attempts. This simplified expression assumes an aborted transaction has no business payoff; Appendix A.1 restores that term and the full comparison between alternatives.

Suppose two approved routes offer a benefit of 5 with 80 percent completion probability. A private proof costing 0.6 leaves 3.4; a signed assessment costing 0.2 leaves 3.8. These values are illustrative.

The private route needs a compensating benefit, such as protecting confidential evidence.

The management lesson is to measure control costs within the workflow. Mandatory checks can ensure that a request meets a rule. They do not make every compliant business arrangement attractive. Contracts, fees, timing and evidence requirements must support the intended cooperation.

Shared limits also need shared enforcement. If two agents each request 60 credits from one 100-credit mandate, each request appears affordable when viewed alone. The protected account must consider both. Adding agents should increase useful capacity without multiplying their principal's spending authority.

4.2 Build first mover advantage through operating capability

Early investment can create a cumulative capability. A team learns which failures matter, builds reusable control interfaces and assembles evidence that future customers can inspect. Where those assets transfer to similar deployments, later integrations may require less repeated work. Strong safety and alignment practices can also support more useful delegation, provided the evidence justifies the expanded scope.

Security and compliance add to this opportunity by protecting the evidence and making obligations traceable to operating controls. The advantage can appear as shorter customer reviews, fewer avoidable interruptions and faster qualification for demanding industrial work. A supplier able to explain and demonstrate its limits gives customers a firmer basis for approval than a broad assurance claim.

These are proposed mechanisms of first mover advantage, not proven financial effects of the three papers. Track qualification time, reuse of validated controls, authorised output and recovery cost. Early entrants also bear integration and standards costs. The durable opportunity comes from accumulated competence and evidence, with the ability to revise both as models and requirements change.

5 Treat safety as an ongoing operating decision

5.1 Permission and renewal under changing conditions

Paper 2 asks a different question: even when an action is authorised, is there sufficient evidence to support what follows? Its core rule is simple. [6]

$$\mathcal{A}^{\text{eligible}} = \mathcal{A}^{\text{mandate}} \cap \mathcal{A}^{\text{assurance}}.$$

In words, **an action is eligible only when it is both permitted and supported by the required assurance**. The intersection symbol means both conditions must hold. A spending mandate does not establish safety. A favourable safety assessment does not confer permission to spend or release data. Both conditions must concern the same action and the same plan for what happens next.

The paper calls that plan a **continuation**. It covers execution, monitoring, fallback, delayed stopping and unfinished remote work. Permission is issued for a bounded period or scope through an **assurance lease**. Renewal requires evidence that still supports the continuing operation.

The full period of potential harm is an **exposure episode**. Paper 2 records each issued lease against one persistent risk allowance for that episode. A replacement worker inherits unfinished work and its history. Restarting the software cannot restore spent allowance. This is a conservative accounting rule; a numerical safety bound also requires the validated models and information assumptions in Appendix A.3.

Shared dependencies matter as much as the number of agents. Two approvals that depend on the same provider or monitor can fail together. Adding workers does not automatically add independent protection.

5.2 The kill switch and verified closure

The **kill switch** is therefore an operating procedure as well as a technical control. It blocks new ordinary work, preserves authorised containment actions, reconciles uncertain outcomes and gathers evidence that the defined exposure has ended. A stop command records an intention. Verified closure records that the required terminal conditions have been established.

This matters in the simulation example. Closing the agent’s session may leave a remote computation running. Cancelling a transfer may leave a file in a queue. The contract must name the party that can stop each effect and the evidence that proves closure. A monetary reservation may be released once its outcome is resolved. Under Paper 2’s construction, an issued risk debit remains charged.

Faster, more reliable containment can support more useful work within a given harm limit. That makes monitoring and recovery infrastructure part of productive capacity, with results that must be tested on the intended deployment.

6 Ask for evidence that supports the exact decision

Paper 3 addresses a common obstacle in business relationships. A customer needs confidence in a supplier’s controls. The supplier needs to protect its internal systems, assessments and commercial information. [7]

The parties can disclose selected evidence, rely on an accepted assessor’s signed statement or use a private proof of an agreed computation. These options have different costs and trust assumptions. A private proof can show that a specified check accepted committed inputs without exposing those inputs. It cannot make a false observation true.

The crucial management question is what the check covers. Three favourable assessments tell an incomplete story if a current suspension has been omitted. Evidence requirements must identify the accepted sources, required categories, relevant time period and adverse records. A proof about yesterday’s configuration also needs a fresh applicability check before today’s action.

Cross-company trust is therefore **claim-specific**. A registry may establish a company’s identity. An assessor may evaluate a defined control or system. A resource owner decides whether those claims support this operation. Trust chaining connects these roles while preserving their scope. A chain of signatures must not turn an identity credential into a safety certificate.

UL Solutions provides a concrete example of why scope matters. Its published Qcells assessment concerns an identified control component in a data-centre energy-management system. It describes documentation review, audits, demonstrations and tests, including approval and recovery controls. A buyer must still connect those findings to the installed component and its conditions of use. [13]

Paper 3 separates the resulting records. The evaluation says the specified check passed. The admission record says current authoritative state accepted the operation. The execution receipt says a resource event occurred. The closure record says the agreed terminal condition was established. Appendix A.4 explains this connection; Appendix A.5 explains the standards and assessment roles.

7 One industrial workflow across all three papers

Return to the manufacturer and its simulation supplier. The following workflow is illustrative, rather than a reported installation. Its value is to show how the three papers change one business process.

Agree one job. The parties specify the design revision, purpose, recipient, permitted output and cost. Paper 1 tests the economics of the arrangement and approved fallbacks. Capacity booking and later data release remain distinct decisions.

Prepare permission and evidence. Paper 2 evaluates the whole planned operation, including shared infrastructure and stopping time. Paper 3 checks the supplier’s scoped evidence against that same job. Before exposure begins, the covering lease is issued and its risk debit is recorded.

Approve at both boundaries. The manufacturer controls release of its design. The supplier controls receipt and processing. Each checks current authority and state. Figure 1 shows how proposing agents connect to these protected controls without owning them.

Handle change without losing history. If the supplier’s assessment is withdrawn, new ordinary work is blocked pending an acceptable resolution. Authorised containment remains available for work already admitted. If an acknowledgement is lost, a retry refers to the original operation. It must not create a second booking or data release.

Close on evidence. The parties establish completion, confirmed cancellation or another agreed terminal outcome. A new agent or model inherits unresolved obligations. The workflow is ready to scale when these failure paths work as reliably as the successful path.

Closure has a precise scope. Stopping processing cannot reverse disclosure of a copied design. Access restrictions, deletion evidence and contractual remedies may address remaining duties, but the original release still requires a decision proportionate to its irreversible consequences.

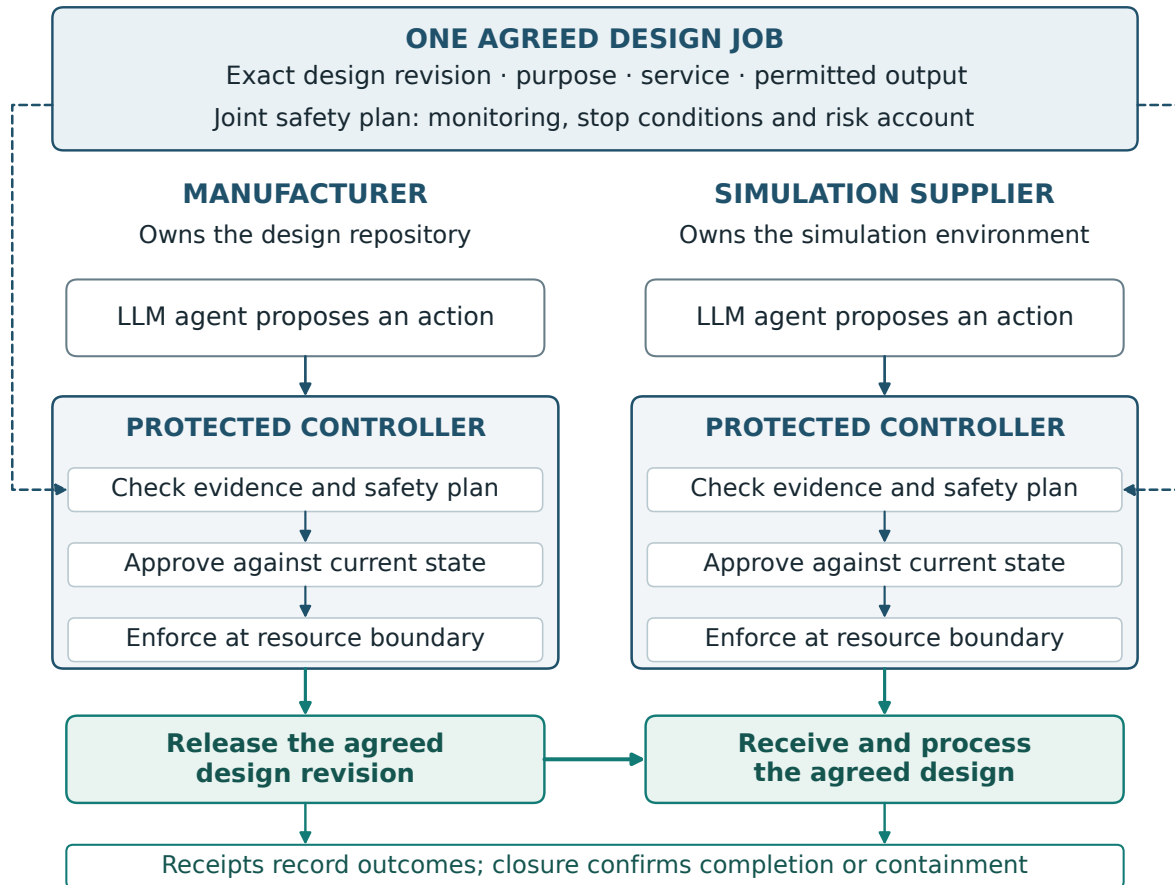


Figure 1: One shared job and two protected resource boundaries. Each LLM proposes; its protected controller checks evidence, approves against current state and enforces at the resource. Dashed lines carry agreed scope and the joint safety plan. Green arrows show controlled effects and outcome records. The joint assurance model must cover both organisations; two local approvals alone do not establish joint safety.

8 Govern an agent swarm from the platform boundary

8.1 One customer mandate and many temporary workers

Consider a provider of generative AI and managed agent services. An industrial customer asks it to assess a software update across several factories. A coordinator assigns temporary agents to review asset records, analyse test results and prepare proposed changes. Workers may request approved computing capacity or supplier evidence. Production changes require separate permission from the resource owner.

This is a proposed design case, grounded in Paper 2’s treatment of worker creation and provider deployments, rather than a validated installation. The provider limits worker types, concurrency, tools and destinations. Open-ended populations and unmodelled tools fall outside that declared scope. (Paper 2, Sections 14.8 and 15.2–15.7) [6]

Each request traces to the customer’s business mandate. Provider service policies impose additional limits. The coordinator and workers can propose actions; the protected controller evaluates evidence, admits an exact operation against current state and enforces it at a covered resource. A peer agent’s message is task data. It cannot grant authority. Agent-generated memory and tool outputs receive the same treatment.

8.2 Keep authority and risk accounts stable as workers change

All workers drawing on the same mandate share its spending limit. Delegating smaller tasks cannot multiply that authority. Business spending and safety risk remain separate accounts. For this example, the assessed harm is release of protected factory records to an unauthorised destination. Production-update safety requires a separate assurance case. The safety account covers the named harm and exposure episode, including relevant interactions, delayed effects and containment. Spawning a worker, replacing a model or retrying a request must preserve issued risk debits, consumed operation slots and pending work. [5] [6]

Figure 2 shows one customer episode inside a platform that serves other tenants. Tenant isolation protects data, credentials and execution. The accounting boundaries follow mandates and exposure episodes: one tenant may contain several mandates, and one mandate may span several services. A limit shared across services needs one authoritative account or justified non-overlapping allocations. Shared models, connectors or monitors remain possible common causes across otherwise isolated tenants.

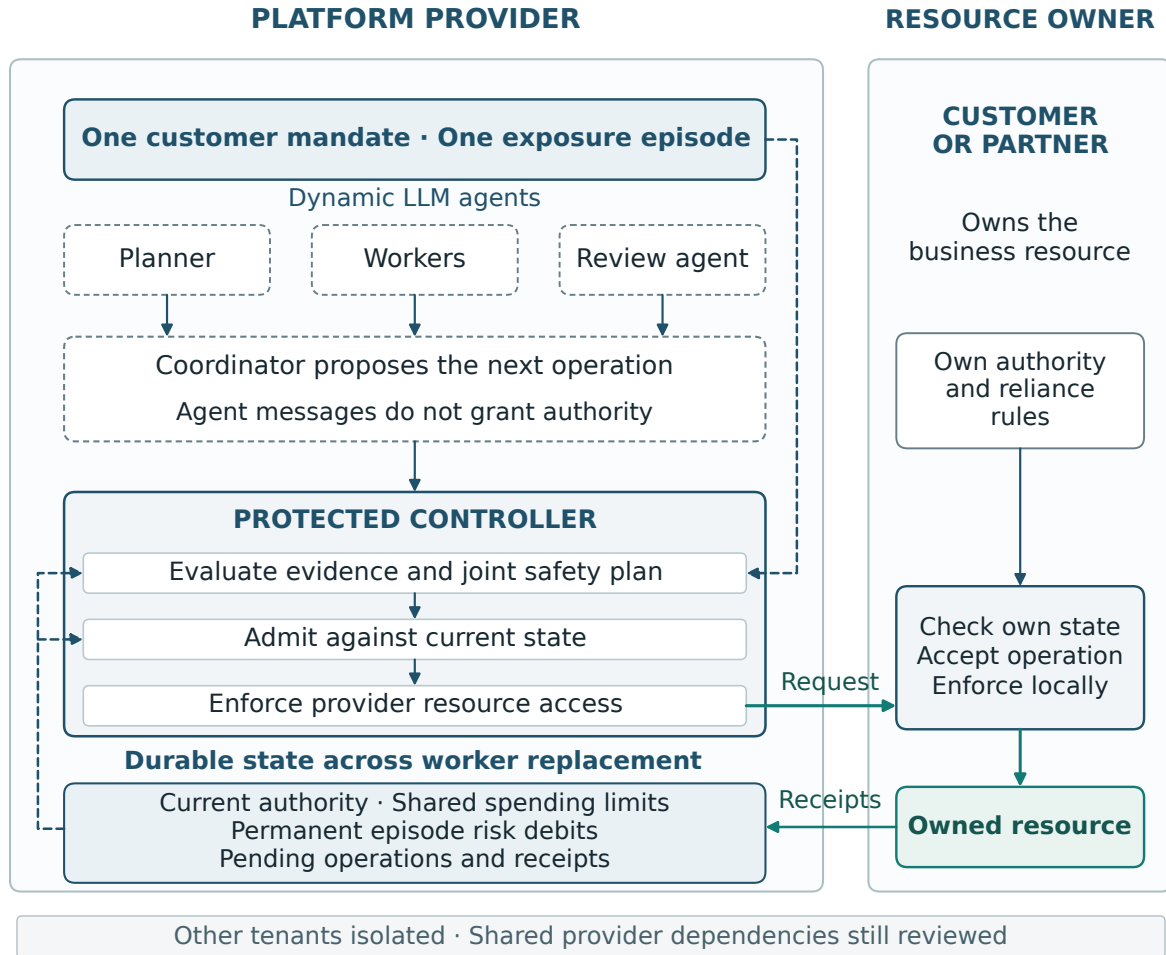


Figure 2: A proposed managed-platform design for one customer episode. Agents propose within a bounded worker pool. Protected controls retain current authority, shared spending, episode risk and pending effects across worker replacement. The provider enforces its own resource boundary; a customer or partner retains local acceptance and enforcement. Outcome records return to protected state. Other tenants remain isolated, while provider-wide incident review addresses shared dependencies. The figure illustrates responsibilities, not a validated platform-wide safety guarantee.

8.3 Follow remote effects through containment and closure

A provider gateway controls provider-operated tools. A customer repository, supplier service or factory controller needs its own enforcement point. Platform admission supplies evidence to that remote decision. Paper 3's records distinguish acceptance, actual execution and closure. A missing acknowledgement remains an unresolved obligation, even when the initiating worker has ended. [7]

Suppose a security alert affects a connector shared by several customers. The provider blocks affected new ordinary work, preserves authorised protective actions and starts agreed containment. Remote owners act at their own boundaries and return evidence. Replacement workers inherit the restriction. Restart follows an authorised review of current evidence and outstanding work. The kill switch must reach these enforcement points; stopping the coordinator alone leaves pending effects unresolved.

The provider should test omitted suspension records, duplicate requests, misleading agent messages and a lost remote receipt. It must also protect the control infrastructure from its own internal coding agents. Changes to admission rules, ledgers and intervention services need separately authorised review. An API-only provider has narrower visibility and control than a managed-agent operator; customer-side tool enforcement remains essential. (Paper 2, Section 15.2) [6]

8.4 Scale only the claims the evidence supports

Paper 1's incentive result concerns specified unilateral deviations. It does not establish resistance to colluding agents or show that cooperation serves a legitimate purpose. Paper 2 requires a coherent assurance model for the joint continuation. Separate tenant risk allowances do not establish a fleet-wide bound when dependencies are shared. Paper 3 requires the implementation's covered effects to preserve the formal rules. Appendix A.2 and Appendix A.3 connect these conditions to the platform case. [5–7]

For the provider, the business opportunity is repeatable customer service with clear limits and evidence about outcomes. For the customer, it is useful parallel work with retained authority over sensitive resources. The next section explains how those responsibilities connect when partners operate under different trust rules.

9 Make cooperation work across trust domains and sectors

A **trust domain** is an organisation or ecosystem with its own authorities, rules and accepted evidence. The manufacturer, simulation supplier and cloud operator may each belong to a different domain. Working across **multiple trust domains** means that a decision depends on several such systems, while each resource owner retains its own authority.

The parties need explicit recognition agreements: which issuer is accepted for which claim, what the claim means, how current it must be and what happens when it is withdrawn. A business identity recognised in another country can establish who the supplier is. Permission to handle a protected design still depends on the receiving organisation's role, mandate and evidence. Shared technical standards make recognition easier to implement; governance defines its scope.

Cross-sector interoperability extends this problem beyond company boundaries. Suppose the simulation supplier's data centre also offers flexible electricity demand to an energy operator. Evidence about its identity, installed assets and available capacity may be useful to both the

manufacturing and energy workflows. The manufacturer’s confidentiality rules and the energy operator’s dispatch and safety rules remain distinct.

The commercial opportunity is to reuse verified evidence through agreed interfaces, reducing repeated onboarding and assessment. The engineering duty is to preserve its meaning and limits. The parties must agree identifiers, units, time periods, permitted uses and responsibility for updates. A capacity statement accepted for simulation planning cannot automatically authorise grid participation.

Successful interoperability therefore has three parts: systems can exchange information, organisations agree what it means, and each has a valid basis for relying on it. The series proposes foundations for that arrangement. Combining sector-specific safety claims still requires a justified joint model of their interactions. Appendix A.5 describes the corresponding trust paths and semantic rules.

10 Understand, govern and monitor the complete system

10.1 Retain the capability to challenge the evidence

Operating competence must extend across every participating trust domain. Organisations can procure models, hosting, assessment and control components. They must still understand how those components behave together, which assumptions support their use and how changing conditions affect permission to act. Accountability for the organisation’s deployment decisions remains with its responsible leaders.

That capability requires people who can inspect supplier claims, test relevant failure paths and interpret operational signals. They need access to deployment versions, configuration, logs, evidence limits and incident information. Where a provider protects internal details, the parties should agree adequate scoped assertions, independent assessment or private verification. If the available evidence cannot support a proposed level of autonomy, leaders should restrict its scope while resolving the gap.

Global providers supply useful components, described below. The comparison identifies responsibilities for the operating organisation. It is neither a provider ranking nor evidence that any product implements the complete research framework.

Table 1 compares provider building blocks and operating responsibilities.

Table 1: Provider building blocks and operating responsibilities

Provider	Documented building blocks	The operating organisation must establish
OpenAI	Tool checks and human review before sensitive actions [14]	Current business authority at every consequential tool
Anthropic	Enforced execution boundaries and persistent session state [15,16]	Continuity of obligations and evidence of remote closure
NVIDIA	OpenShell runtime controls and NemoClaw deployment blueprints [17,18]	Policy settings bound to the exact operation and deployment
DeepSeek	Efficient model designs and downloadable weights [19,20]	Secure operation and independent control of business effects

Table 1 (continued)

Provider	Documented building blocks	The operating organisation must establish
Moonshot	Downloadable models trained for tools and agent tasks [21, 22]	Delegation limits and durable records of completed and pending work
Kimi		
Alibaba	Qwen models and AgentRun hosting, sandbox and identity services [23–25]	The business transactions permitted by each technical credential

The published DeepSeek V3, Kimi K2 and Qwen3 releases are examples of inspectable deployment choices, rather than claims about the latest model rankings. The relevant comparison is the complete operating arrangement: who runs the model, who can change it, who sees the evidence and who controls the tools.

10.2 Allocate responsibilities across deployment choices

With a **managed API**, the provider operates inference and the customer integrates it into business systems. This can reduce initial operating work. Contracts and controls must address version changes, data handling, evidence access, incident response and cancellation behaviour.

With **independently operated open weights**, the organisation gains more direct control over model deployment. It also takes on serving, security, updates and evaluation. Access to weights does not necessarily include training data, reproducible training or unrestricted licensing. Changes to quantisation, tools or hosting can require renewed evidence.

A **hybrid arrangement** can allocate work across both. It adds routing and recovery responsibilities. A fallback model must satisfy the relevant conditions before inheriting a task; compatibility with an API alone does not establish that permission.

Keep an inventory linking each model and tool to its owner, permitted effects, evidence and change triggers. Recheck material changes to models, prompts, tools, data sources and runtime configuration. Monitor failed controls as well as model quality. Contracts should provide workable notification, investigation and exit routes. A supplier certificate is an input to this process, rather than a substitute for it.

Across all three choices, measure total workflow cost per completed, authorised and quality-accepted outcome. Include inference, checking, retries, human review, containment and recovery. A lower token price may help. Its commercial value depends on how much acceptable work the whole system produces.

11 Build sovereignty through operational choices

The architecture supports a globally useful view of sovereignty: the ability to choose suppliers, govern operations, inspect relevant evidence and continue or exit when conditions change. Control over model weights, operating infrastructure and business authority are separate decisions.

North America offers major model platforms, infrastructure and enterprise ecosystems. Buyers can use that scale while retaining independent control of consequential tools and business records. Portable evidence and tested migration paths can reduce dependence on one operating arrangement. Multiple suppliers improve resilience only where their critical dependencies also differ.

China and the wider APAC region provide important examples of model efficiency, open-weight access and rapid iteration. These expand options for local operation and industrial integration. Fast release cycles increase the need to identify exactly which version was assessed and deployed. APAC also includes distinct governance approaches, such as Singapore’s agentic AI framework; regional requirements should be evaluated in their own context. [26]

Europe can build trusted AI capability through three routes, combined where the workflow requires it. Each offers a different balance of capability, operating responsibility and dependence.

Use US AI platform services. Companies can draw on established models and managed infrastructure while retaining control over their own business resources. They must understand the provider’s limits, monitor changes and place consequential tool use behind enforceable permissions. Contracts should support evidence access, incident response and a tested exit route, as discussed in Section 10.2.

Operate open-source or open-weight models under sovereign control. This includes models developed in China, such as DeepSeek and Qwen. Adoption already extends to European industry: Siemens told Reuters in June 2026 that it uses both within a wider model portfolio. [27] Local operation can give the company control over deployment, data flows and updates, subject to the model’s licence and infrastructure dependencies. The operating team must assess the model and its software supply chain, restrict network access and control telemetry. Model origin and local hosting alone establish neither safety nor alignment.

Build on European foundation and sector-specific models. Mistral provides current options for running models in customers’ own environments and choosing an inference region for its managed service. The contractual scope and any permitted transfers still need review. [28] IPCEI-AI—the Important Project of Common European Interest in Artificial Intelligence—offers a complementary route to developing European capability. Its intended scope includes foundation and sector-specific models, industrial data and deployment services. Nineteen Member States jointly designed the initiative; eleven announced plans to start State-aid pre-notification in September 2026. Leaders should treat it as a development programme entering assessment, rather than an already delivered model service. [29, 30]

Design the controls from day one. Across all three routes, leaders must fund teams that understand, design, implement and monitor AI safety and alignment controls alongside the model deployment. The three papers provide a common foundation: protected authority, ongoing assurance and verifiable evidence across trust domains. [5–7]

Keep the LLM agent in the proposer role. A deterministic controller should admit actions only when the mandate, current evidence and operating limits permit them. Enforce that decision at the resource itself. Containment must restrict reachable tools, data and networks; the kill switch must block new ordinary actions, preserve authorised protective functions and start the agreed containment of pending work. Preserve unresolved obligations and risk accounting when changing a model or supplier. Renew the deployment evidence and assurance before resuming work. Appendix A.2 and A.3 explain the assumptions behind these safeguards.

Apply cryptography from the start to bind identity, authorisation and provenance into a verifiable chain. Signed credentials and protected records should connect the organisation’s mandate to the agent, assessed deployment configuration, approved action and execution receipt. That configuration includes the model, tools and controller. Protect the signing keys and check expiry and revocation. A valid signature establishes authenticity and integrity; it does not establish that a claim is true or an action safe. Appendix A.4 explains how private verification can support specific eligibility checks without disclosing all underlying evidence.

Within a company, these controls connect business owners to the resources for which they are accountable. Across companies, each trust domain retains authority over its own resources.

Partners must agree what evidence means, how permissions are withdrawn and who closes pending work when a provider changes or fails. This supports the cross-sector interoperability described in Section 9 while preserving local responsibility.

European Business Wallets can support organisational identity and delegation within this wider architecture. For providers such as Spherity, the opportunity is to connect those credentials to current evidence and resource decisions across all three deployment routes. This is a proposed implementation role, not a completed deployment claim.

Regulation creates value when it is translated into usable controls and credible evidence. NIST’s voluntary risk-management framework, the EU AI Act and Singapore’s governance framework have different legal roles. They provide reasons to address responsibility, evaluation and intervention early, while deployment-specific obligations remain a matter for the relevant jurisdiction and use case. [9, 26, 31]

For policymakers, the practical agenda is evidence portability, competent assessment and clear responsibility across borders. For entrepreneurs, it includes services that make those controls cheaper to integrate and easier to test. For researchers, it requires proving that the proposed guarantees survive real deployment conditions.

12 Give the operating model an owner before expanding autonomy

Begin with one workflow whose resources and closing conditions can be specified. A controlled design release, compute booking or bounded procurement step offers a clearer starting point than unrestricted delegation. Safety belongs in that initial design, because permissions and resource access shape which failures can later be contained.

The following decisions should precede expansion.

Table 2 sets out the decisions and evidence required before expansion.

Table 2: Decisions and evidence required before expanding autonomy

Accountable owner	Decision to make	Evidence required before expansion
Business executive	Which outcome justifies delegation and which alternatives are allowed	Economics including checks, failed attempts and recovery
CTO and security lead	Where authority is enforced and who controls each resource	Tests of bypass, replay, concurrent requests and recovery
Safety lead	Which harms matter and how unfinished work is contained	Deployment evidence, monitoring limits and stop-to-closure tests
Legal and compliance lead	Which obligations and assessments apply	Claim scope, competent issuers and deployment applicability
Procurement and operations	Who handles changes, incidents and supplier exit	Version commitments, escalation ownership and a migration exercise

Test an expired permission, omitted suspension, changed recipient and duplicate request. Withdraw a supplier claim while work is pending. Lose an acknowledgement. Replace an agent

and restart the controller. The expected result is both useful authorised work and preserved responsibility when the process fails.

Report a small set of operating measures: accepted business outcomes, total cost per outcome, time to block new admissions, time to verified closure and unresolved work by age and owner. Where numerical risk bounds are claimed, report their calibration and assumptions separately. Appendix A.6 records what the papers have actually tested and which implementation gaps remain.

Before delegating more authority, the responsible executive should be able to trace an actual operation to its mandate, current evidence, enforcement record and remaining obligations. That trace is a concrete basis for deciding where autonomy can expand.

13 Outlook and a practical recipe for leaders

The next opportunity is to make justified autonomy repeatable across plants, suppliers and sectors. That requires evidence and enforcement that remain meaningful when components change. NIST’s work on an AI risk-management profile for critical infrastructure illustrates this growing focus on industrial operating conditions; the profile remains under development. [32]

First, assign an accountable executive and one bounded use case. Define the useful outcome, affected people, permitted effects and conditions for stopping. Include the partner that controls each remote resource.

Second, fund a standing interdisciplinary team. Combine AI and formal-methods expertise with domain and control engineering, cybersecurity, legal and compliance work, human factors and operations. Give it time, test facilities, access to evidence and authority to challenge deployment. This collaboration is a prerequisite for the approach proposed here, rather than an optional review at the end.

Third, invest in the shared operating infrastructure. Establish identity and delegation, protected admission, monitoring, persistent records and tested containment. Reserve funding for evaluation, maintenance and incident exercises alongside inference and product development.

Fourth, test the safety and security assumptions against the workflow. Simulate faults, delays and shared failures. Challenge the model with deployment data. Use independent review for consequential claims and retain validated fallback and containment responses for specified failure conditions.

Fifth, expand quickly against evidence and report the results. Review useful authorised work, customer qualification time, control failures, containment and remaining obligations. Share relevant incident lessons across the ecosystem. Investors and society should be able to see what autonomy delivers, how it is governed and who can intervene.

AI safety and alignment are foundational obligations of responsible leadership. Unsafe or misaligned AI can have catastrophic consequences, including loss of life, disruption of critical infrastructure and failures that spread across companies and public institutions. Leaders who adopt or deploy AI must make safety, alignment, security and applicable compliance conditions of deployment from the outset. They must understand the limits of the evidence, fund effective controls and retain the ability to intervene. This responsibility continues throughout operation and across delegated activities.

Act before granting AI authority. Assign an accountable executive, fund a standing interdisciplinary team and require a documented safety and alignment assessment for each consequential use case. Make deployment conditional on evidence that controls work under the defined conditions. Test containment, shutdown and recovery before expanding autonomy. Monitor changes and suspend or restrict ordinary AI-directed operations when safety conditions fail, while preserving authorised protective functions. These are minimum requirements of responsible AI risk management. Every material delegation of authority must carry clear responsibility, enforceable limits and a tested path to intervention.

A Theoretical foundations

This appendix explains the research behind the management decisions. It preserves the distinction between incentive results, authority guarantees, conditional safety bounds and implementation proposals. The source manuscripts contain the full definitions and proofs.

Table 3 maps the main text to its formal foundations.

Table 3: Connections between the main text and theoretical appendix

Main text passage	Technical connection	Appendix
Make the economics of control part of the business case	Completion, costs and permitted deviations	A.1
Design the controller and govern the platform swarm	Controller invariants, worker creation and persistent state	A.2
Govern ongoing safety and shared platform dependencies	Joint continuations, leases, episode risk and aggregation limits	A.3
Ask for evidence and follow the industrial workflow	Private verification and execution refinement	A.4
Cooperate across trust domains and sectors	Standards, trust paths and claim-specific reliance	A.5
Give the operating model an owner	Reported evidence and remaining research	A.6

A.1 The economics behind the simple value formula

Paper 1 models a finite game in which principals choose bounded programs that may respond to permitted information about counterpart programs. For a full program profile p , implemented utility is

$$V_i(p) = a_i + q(p) (u_i(p) - a_i) - c_i(p).$$

Here, $u_i(p)$ is the payoff while authorised, a_i the common abort payoff, $q(p)$ the probability of completing before revocation and $c_i(p)$ expected implementation cost. Setting $a_i = 0$ gives the simple formula in the main text. Costs include attempts that fail to finish. (Paper 1, Equation 3) [5]

For a candidate profile p^* and a unilateral deviation d_i , with resulting profile p^d , the exact incentive margin is

$$M_i(d_i) = q^* g_i(d_i) - (q_d - q^*) (u_i(p^d) - a_i) - \Delta c_i(d_i).$$

The active payoff advantage is $g_i(d_i) = u_i(p^*) - u_i(p^d)$; $q^* = q(p^*)$ and $q_d = q(p^d)$. The cost saving is $\Delta c_i(d_i) = c_i(p^*) - c_i(p^d)$. The terms separate active advantage, timing advantage and cost saving. The candidate is a Nash equilibrium in the declared finite game exactly when every permitted unilateral deviation has a non-negative margin. (Paper 1, Equations 5–6) [5]

This result supports the business comparison of evidence routes. It does not predict that an LLM will optimise the specified utility or establish that the game’s incentives are socially

desirable. The paper’s private-interface result also limits what a counterparty can infer from indistinguishable observations. Confidential verification must expose the operational distinctions necessary for a valid decision.

A.2 What the controller guarantees

Paper 1 separates a narrow evaluator from protected stateful execution. Its controller maintains current authority, delegation, configuration, shared reservations and consumed identifiers. Under sound checking, complete mediation and atomic execution, the covered digital commit preserves the stated authority, budget, exact-action, delegation and replay invariants. Complete mediation means every covered effect passes through the protected boundary.

Paper 2 adds an assurance supervisor for the joint continuation and persistent episode account. Paper 3 refines the implementation into pure evaluation, protected admission and resource-side enforcement. An evaluator result is insufficient until the current state service accepts it and the resource adapter enforces the accepted operation.

There is a material difference between Paper 1’s atomic resource commit and Paper 3’s local admission before a remote effect. Asynchronous execution needs durable pending records, protected operation slots, receiver-side idempotency and reconciliation. Idempotency means repeating delivery of the same operation does not commit its covered effect twice. A caller-selected new identifier cannot create a new entitlement.

The control plane supplies current authority and assurance decisions. The data plane carries execution and its observations. Their convergence is the binding of each effect to those decisions, with authenticated feedback about progress. It is not a merger of agent-writable data with protected policy state. Logical separation can be preserved even when components share a deployment.

Operational alignment is conformance to the declared mandate and supported continuation. Behavioural evaluations and broad human-value alignment are separate subjects. The controller’s determinism provides repeatable transitions over complete inputs; uncertainty in observations, policy adequacy and the wider environment remains.

The platform case in Section 8 uses Paper 2’s bounded worker-creation rules. Worker templates, delegated scope and deployment conditions must be checked before admission. Controller state remains persistent across replacement. A tenant identifier does not define a principal or an exposure episode; implementations must bind those objects explicitly. (Paper 2, Sections 14.8 and 15.2) [6]

A.3 The safety conditions behind permission and renewal

The main text suppresses the time index in Paper 2’s eligibility rule. The full shorthand is

$$\mathcal{A}_t^{\text{eligible}} = \mathcal{A}_t^{\text{mandate}} \cap \mathcal{A}_t^{\text{assurance}}.$$

Both sets must concern the same exact action and compatible continuation. The construction first retains the full continuation and then projects to its initial action. Separate local permissions cannot establish joint safety without a coherent assurance model covering interactions and shared dependencies. (Paper 2, Equations 9–10) [6]

Paper 2 permanently debits each issued continuation against the episode allowance:

$$B_{n+1} = B_n - \alpha_n, \quad B_0 = \varepsilon, \quad \alpha_n \leq B_n.$$

Here, α_n bounds the new lease's incremental first-harm probability through closure; B_n is remaining allowance and ε the initial allowance. Under the theorem's assumptions,

$$\Pr(\tau_H \text{ lies in the episode}) \leq \varepsilon + \delta.$$

The calibration coverage error is δ ; τ_H is the first occurrence of the named harm. Assumptions include a containing finite model bank, sound inference, initially absent harm, mediated execution, covered containment and the restriction on information used to select proposals. Unobserved prior harm must remain represented. The bound does not follow from bookkeeping alone and does not bound every measure of severity or repeated harm. (Paper 2, Equations 7–8) [6]

In particular, undisclosed information used to choose proposals can invalidate a calculation conditioned only on declared history. Privacy from a counterparty must not omit information needed by the logical assurance evaluator. Renewal must also carry pending effects forward. Cancellation does not refund an issued debit. A business reservation has separate contractual release rules.

For the kill switch, distinguish blocking new ordinary admission, effective containment, verified closure and controlled restart. Authorised protective actions must remain possible. A late effect can occur after revocation when it was admitted earlier. Such outcomes remain within the continuing containment and accounting obligations.

For the platform swarm, the joint continuation must represent shared connectors, monitors and other relevant common causes. A fleet-level probability claim requires its own population, harm definition and justified aggregation argument. The episode theorem alone does not supply that claim. (Paper 2, Sections 15.2–15.7) [6]

A.4 Private evidence and the connection to execution

Paper 3 proposes the private relation

$$R(x, w) = \text{Src} \wedge \text{View} \wedge \text{Sem} \wedge \text{Mand} \wedge \text{Assur} \wedge \text{Bind}.$$

The public statement x binds the operation, continuation, versions, evidence roots, audience and validity context. The private witness w contains protected records and the evaluation trace. The predicates check accepted sources, the required complete view, agreed semantics, mandate, assurance and exact context binding. Completeness is relative to declared sources and queries, including adverse records; it cannot establish completeness about the world. (Paper 3, Equation 8) [7]

One assurance profile verifies an accepted service's authenticated assertion. Another verifies the specified computation itself. The first relies on the service for computational correctness. Both still depend on source provenance and the adequacy of the underlying model. A proof prepared before lease issuance must bind the exact proposed state transition and expected predecessor; its acceptance receipt identifies the transition actually recorded.

The execution target is conditional refinement:

$$\alpha(\text{Traces}_{\text{fin}}(C)) \subseteq \text{Traces}_{\text{fin}}(A).$$

Here, C is a concrete implementation, A the abstract controller and α the abstraction map. This α is unrelated to the lease debit α_n in A.3. Concrete events must map to permitted abstract transitions or genuine state-preserving steps, with covered effects retained. (Paper 3, Equation 16) [7]

Trace refinement can preserve authority and state invariants. Transferring Paper 2’s numerical harm bound additionally requires preservation or conservative bounding of probability, observation, timing and proposal-selection laws. Waiting for proofs or messages can increase exposure even if logical state is unchanged. This distinction is central to the industrial workflow’s implementation obligations.

A.5 Standards and claims across organisations

Agent2Agent (A2A) provides message, task and extension mechanisms. Paper 3 uses version 1.0.0 as a reviewed baseline, while noting later releases. Its proposed application profile supplies exact-operation bindings, evidence challenges and typed records. Task completion and message delivery do not themselves establish resource closure. Implementations must pin the protocol and schema versions. [33] [7]

W3C Verifiable Credentials can carry attributed identity, delegation and assessment claims. Open Digital Rights Language (ODRL) can express permissions, prohibitions and duties. Paper 3 requires a bounded policy profile with precise rules for missing evidence, conflicts, units, mappings and future duties. RDF representations, SHACL structure checks and PROV-O provenance can support evidence processing. They do not by themselves establish agreement about business meaning. [34, 35] [7]

Claim validation distinguishes operational alignment, safety, regulatory compliance, standards conformity and control effectiveness. It checks the assessed object, deployment, method, issuer competence, jurisdiction, role, validity, status and dependencies. The proposed outcomes are Support, Reject, Indeterminate and Error. Support means that the specified reliance rule is satisfied. Required unresolved or erroneous claims cannot authorise ordinary admission.

Trust paths separately establish identity, mandate, evidence authority and assurance compatibility. Each delegation or semantic mapping must preserve or narrow accepted scope. Management-system certification, technical testing, cryptographic verification and regulatory determinations answer different questions. A system-level conclusion needs a justified composition rule covering interactions and common causes.

Across sectors, a mapping must preserve subject identity, claim type, units, temporal scope and issuer authority. Converting a duration from minutes to seconds can be a bounded transformation. Treating a manufacturing approval as an energy-market authorisation changes meaning and requires separate authority. Reciprocal recognition may therefore be asymmetric and specific to a use case. Syntactic compatibility alone establishes neither shared semantics nor permission to rely.

A.6 Evidence and the next research gates

The following counts report results in the reviewed manuscripts. They describe different artifacts and must not be added into a combined assurance score. They were not rerun for this editorial revision.

Table 4 records the reported checks and remaining validation needs.

Table 4: Reported research checks and remaining validation needs

Paper	Reported method and result	Remaining validation need
1	Exact derivations; 38,880 rational checks; controllers with 1,427 and 1,819 states; five control-omission counterexamples	Richer deviations, empirical workflow costs and deployed boundary validation
2	Conditional renewal theorem; 432 synthetic scenarios; 120,060 terminal histories; maximum computed harm 3.73% under a 5% test threshold; controller up to 24,448 states	Field calibration, joint dependency modelling and implementation assumptions
3	Five conditional propositions; 16,020 states and 60,238 transitions; nine invariants; eight control omissions; 29 reference-gate scenarios	Complete private proof implementation, privacy and performance measurement, distributed refinement

Paper 1’s checks support the declared games and mediated digital effects. Paper 2’s results concern synthetic settings, rather than measured field risk. Paper 3’s finite abstraction and reference gate do not implement a complete cryptographic protocol or validate an OpenClaw deployment. The series review record separately identifies reproduced checks and unavailable companion-package reruns.

Deployment claims require an implemented workflow with calibrated observations, measured control costs and realistic concurrency. Testing must cover useful authorised work, rejection, remote containment, recovery and verified closure.

References and source notes

The three source manuscripts are the reviewed versions dated 28 September 2026. Public sources support the attributed findings and documented mechanisms. Strategic advantages are identified as inferences, and survey responses are distinguished from observed performance. All online sources were accessed on 28 September 2026. The abbreviation n.d. indicates that a reliable publication date is unavailable.

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